

```

1: (θfunc, θpass)
2: Prgm
3: DispG
4: misc\statline("msg:▪ Initializing...")
5:
6: string(approx(θfunc))→θf
7: expr(left(θf, inString(θf, "=")-1) & "-" & right(θf, dim(θf)-inString(θf,
  "=")) & ")→θif")
8:
9: Local θi, θpass, θΔx, θΔy, θlr, θtr, θxp, θyp, θp1, θp2, θp3, θp4, θymin, θymax, θyl
10:
11: For θi, 0, 76
12:   StoPic #("p"&string(θi)), θi, 0, 159, 1
13: EndFor
14:
15: While true
16:   misc\statline(&string(ln(θpass)/ln(2)) & "...", "msg:♦ Initializing detail level
    2^")
17:
18:   (xmax-xmin)/(2*θpass)→θΔx
19:   (ymax-ymin)/θpass→θΔy
20:   newList(2*θpass+1)→θlr
21:   newList(2*θpass+1)→θtr
22:   xmin-Δx/2→x
23:   ymax+Δy/2→y
24:
25:   For θxp, 1, 2*θpass+1
26:     LineVert x, -1
27:     θif≥0→θlr[θxp]
28:     LineVert x, -1
29:     x+θΔx→x
30:   EndFor
31:   xmin-Δx/2→x
32:   y-θΔy→y
33:
34:   For θyp, 2, θpass+1
35:     misc\apdreset()
36:     misc\statline(&string(ln(θpass)/ln(2)) & "...
    "&string(round(100*(θyp/(θpass+1)), 1)) & "%", "msg:♦ Rendering detail level 2^")
37:
38:     exact(round((ymax+Δy/2-(y+θΔy))/Δy, 0))→θymin
39:     exact(round((ymax+Δy/2-y)/Δy, 0))→θymax
40:     seq(θi, θi, θymin, θymax)→θyl
41:     For θi, 1, dim(θyl)
42:       RplcPic #("p"&string(θyl[θi])), θyl[θi], 0
43:     EndFor
44:
45:     θlr[1]→θp1
46:     θif≥0→θp3
47:     θp3→θtr[1]
48:
49:     For θxp, 2, 2*θpass+1
50:       x+θΔx→x
51:       θlr[θxp]→θp2
52:       θif≥0→θp4
53:
54:       If θp1≠θp2 or θp3≠θp4 or θp1≠θp3 or θp2≠θp4 Then
55:         {impprgm\drawrect(x-θΔx, y+θΔy, x, y)
56:       EndIf
57:
58:       θp4→θtr[θxp]

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59:     $\theta p2 \rightarrow \theta p1$ 
60:     $\theta p4 \rightarrow \theta p3$ 
61:  EndFor
62:
63:     $\theta tr \rightarrow \theta lr$ 
64:     $xmin - \Delta x / 2 \rightarrow x$ 
65:     $y - \theta \Delta y \rightarrow y$ 
66:  EndFor
67:
68:     $2 * \theta pass \rightarrow \theta pass$ 
69: EndWhile
70:
71: EndPrgm
```